

Chapter 10

Magnetism III: Magnetic Symmetry and Magnetic Structures

Abstract We discuss magnetic symmetry and determination of magnetic structures in this chapter. The antiparallel alignment of spins in antiferromagnetic and ferrimagnetic crystals is not always linear. The spins assume a variety of orientations resulting in exotic magnetic structures. The chapter begins with the symmetry aspects of spin ordering. The concepts of magnetic symmetry, magnetic symmetry elements, magnetic point groups and magnetic space groups are explained. The preferred tool for the experimental determination of magnetic structures is neutron diffraction. Neutron diffraction theory is discussed and neutron diffraction instrumentation is described in detail. Neutron diffraction studies started with the antiferromagnetic manganese oxide and manganese fluoride. Studies of more complicated magnetic structures are also discussed. An important class of magnetic structures is the helical structure. Its stability is discussed. Magnetic structures of rare earth elements are described.

10.1 Introduction

We have seen in the preceding chapter that in some materials, the spin magnetic moments assume ordered positions in a certain temperature range. There are three types of ordering: (i) equal parallel spins, (ii) equal antiparallel spins and (iii) unequal antiparallel spins. These spin arrangements are shown in Fig. 10.1. These orderings give rise to three types of magnetic materials: ferromagnetic, antiferromagnetic and ferrimagnetic. This is only a broad classification. Within the latter two groups, the spins assume a variety of orientations. As an introduction to the exotic field of magnetic structures, several spin arrangements are shown in Figs. 10.2 and 10.3.

In this chapter we introduce magnetic symmetry, magnetic point groups and magnetic space groups. The neutron diffraction method of determining magnetic structures is described in detail. Several examples of solved magnetic structures are discussed.

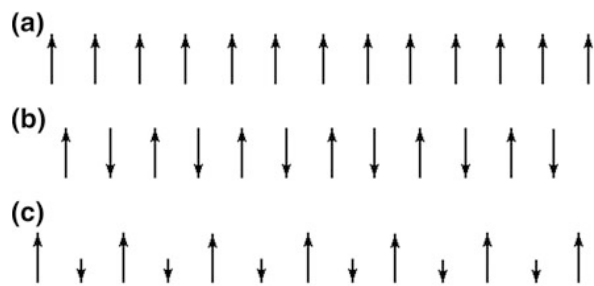


Fig. 10.1 Linear arrays of spins illustrating **a** Ferromagnetic, **b** Antiferromagnetic and **c** Ferrimagnetic orderings

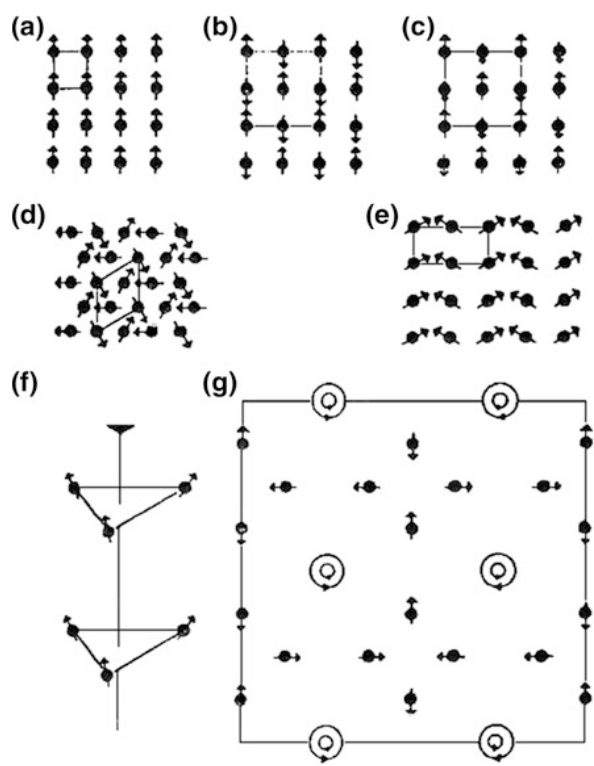
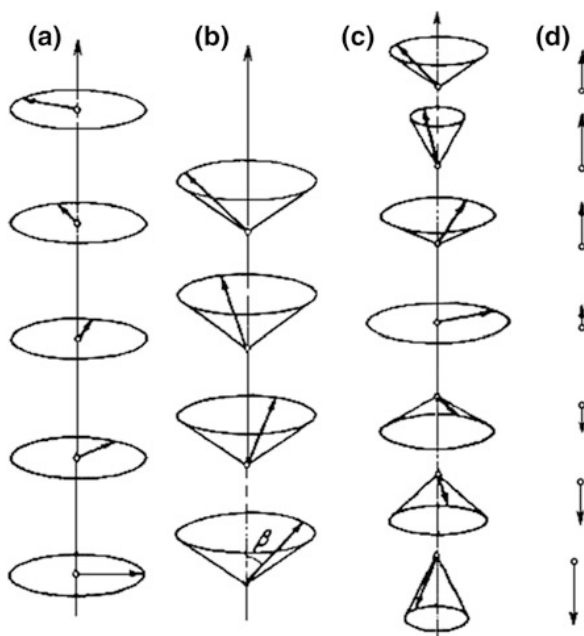


Fig. 10.2 Types of magnetic structures: **a** Ferromagnet, **b** Antiferromagnet, **c** Ferrimagnet, **d** Triangular structure, **e** Weak ferromagnet, **f** Umbrella structure, **g** Three dimensional (multiaxial) structure

The topic of magnetic symmetry is treated by Bhagavantam [1], Prandl [2], Cracknell [3] and Sirotnin and Shaskalskaya [4]. Oles et al. [5] have compiled solved magnetic structures.

Fig. 10.3 Types of helicoidal structures: **a** Simple spiral, **b** Ferromagnetic (*conical*) spiral, **c** Complex spiral, **d** Static longitudinal spin wave



10.2 Magnetic Symmetry

10.2.1 General

A detailed treatment of magnetic symmetry involves use of group theory which is beyond our scope. We shall therefore develop the concept of magnetic symmetry qualitatively and only note the results of rigorous treatments.

Before taking up magnetic symmetry, let us briefly consider the principles of conventional crystal symmetry. A crystal has symmetry elements inversion, rotation axes, mirror planes and rotation-reflection axes. Independent combinations of these symmetry elements are called point groups; there are 32 point groups. Crystals (assumed to be infinite in extent) also have translational symmetry elements, glide planes and screw axes. Independent combinations of point group symmetry elements and translational symmetry elements are called space groups; there are 230 space groups.

Some crystals contain atoms with non-zero spin. These atoms have an associated spin magnetic moment. If these magnetic moments are ordered in a definite manner, the crystal will have additional symmetry elements. Following Bhagavantham [1], we shall consider as an example the system shown in Fig. 10.4. The arrows indicate the sense of spins. In Fig. 10.4a, if the spins are ignored, the system has X_3 as a 2-fold rotation axis and X_2X_3 as a mirror plane. But if the spins are considered, X_2X_3

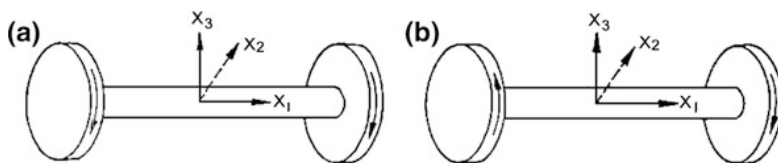


Fig. 10.4 Illustration of magnetic symmetry operation; **a** Spins down, **b** Antiparallel spins

is still a reflection plane but X_3 is no longer a rotation axis. Again, in Fig. 10.4b, if spins are ignored, X_3 is a rotation axis and X_2X_3 is a reflecting plane. But in this case, if the spins are considered X_3 is still a rotation axis but X_2X_3 is not a reflection plane. The important conclusion is that the symmetry elements of a system with spins are different from those of a system without spins.

In the case discussed above, on rotation (in Fig. 10.4a) and on reflection (in Fig. 10.4b), the up-spins become down-spins and vice versa. The system can be brought into congruence with itself by once again reversing the spins, up to down and down to up. This composite operation is called complementary operation and is denoted by symbol R in the Zheludev notation. Shubnikov, on the other hand, introduced magnetic symmetry in a different way. He regarded objects as having elements with black and white colours. An operation that changes black to white and white to black is equivalent to a symmetry operation which reverses spins. The absence of a change in colour is indicated by gray colour. In Shubnikov's system, the operation is denoted by $1'$. In the Shubnikov notation, $1'$ is called the 'time inversion' operator. Applying $1'$ twice will reproduce the original colours.

10.2.2 Magnetic Point Groups

The principle of deriving magnetic point groups is to combine the symmetric operator $1'$ with the ordinary (conventional) point group symmetry operations. The result is a magnetic point group. Thus, addition of $1'$ to the point group 4 results in magnetic point groups 4, $4'$ and $4 \times A$; A is antisymmetric group comprising operations 1 and $1'$. These are shown in Fig. 10.5. Similarly, point group m results

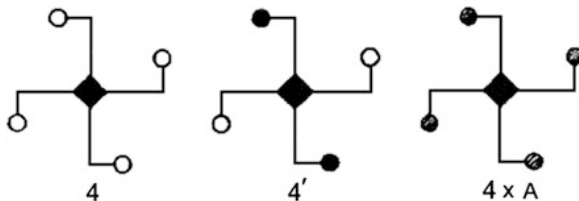


Fig. 10.5 Operation $1'$ acting on point group 4



Fig. 10.6 Operation $1'$ acting on point group m

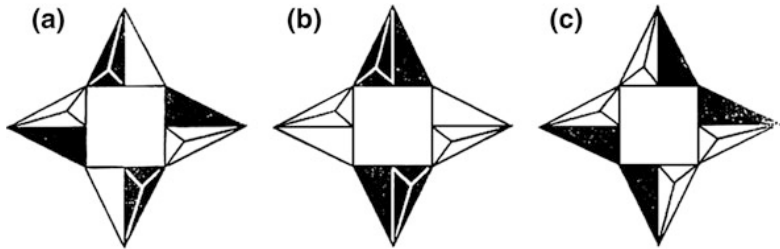


Fig. 10.7 Operation $1'$ acting on point group $\bar{4}2m$; **a** $\bar{4}2'm'$, **b** $\bar{4}'2m'$ and **c** $\bar{4}'2'm$

in magnetic point groups m and m' (Fig. 10.6). Sirotin and Shaskalskaya [4] have given the example of conventional point group $\bar{4}2m$ which on application of $1'$ results in $\bar{4}2'm'$, $\bar{4}'2m'$ and $\bar{4}'2'm$ (Fig. 10.7). The total number of magnetic point groups is 90 as against the 32 conventional point groups. A complete list of magnetic point groups is given in Table 10.1.

10.2.3 Magnetic Space Groups

The combination of conventional space group symmetry elements with the magnetic symmetry element $1'$ results in magnetic space groups. The magnetic space groups were first derived and listed by Belov et al. [6]. They are 1421 in number as against the 230 conventional space groups. This means that each conventional space group has several magnetic space groups associated with it. These are listed in Table 10.2 for the space group $D_{2h}^{17}/Cmcm$. The magnetic space groups of some magnetic crystals are given in Table 10.3.

Before proceeding further, we shall summarize the results of conventional (or ordinary) crystallography and magnetic crystallography (Table 10.4).

Table 10.1 List of the 90 magnetic point groups

1					
$\bar{1}$	$\bar{1}'$				
2	2'				
m	m'				
2/m	2'/m	2/m'	2'/m'		
222	2'2'2'				
$mm2$	$m'm'2'$	$m'm'2$			
mmm	$m'm'm$	$m'm'm$	$m'm'm'$		
4	4'				
$\bar{4}$	$\bar{4}'$				
4/m	4'/m	4/m'	4'/m'		
422	4'22'	42'2'			
4mm	4'm'm	4 m'm'			
$\bar{4}2m$	$\bar{4}'2'm$	$\bar{4}'2m'$	$\bar{4}2'm'$		
4/mmm	4/m'm'm	4'/m'm'm	4'/m'm'm	4/m'm'm'	4/m'm'm'
3					
$\bar{3}$	$\bar{3}'$				
32	32'				
3m	3m'				
$\bar{3}m$	$\bar{3}'m$	$\bar{3}'m'$	$\bar{3}m'$		
6	6'				
$\bar{6}$	$\bar{6}'$				
6/m	6'/m	6/m'	6'/m'		
622	6'2'2'	62'2'			
6mm	6'm'm	6m'm'			
$\bar{6}m2$	$\bar{6}'m'2$	$\bar{6}'m'2'$	$\bar{6}m'2'$		
6/mmm	6/m'm'm	6'/m'm'm	6'/m'm'm	6/m'm'm'	6/m'm'm'
23					
$m\bar{3}$	$m'3$				
432	4'32'				
$\bar{4}3m$	$\bar{4}3'm'$				
$m\bar{3}m$	$m'3m$	$m3m'$	$m'3m'$		

Table 10.2 The family of magnetic space groups belonging to $D_{2h}^{17}/Cmcm$

$Cmcm$	$C_p mcm$
$Cm'cm$	$C_p m'cm$
$Cmc'm$	$C_p mc'm$
$Cmcm'$	$C_p mcm'$
$Cm'c'm$	$C_p m'c'm$
$Cmc'm'$	$C_p mc'm'$
$Cm'cm'$	$C_p m'cm'$
$Cm'c'm'$	$C_p m'c'm'$

Table 10.3 Chemical and magnetic space groups of some crystals

Crystal	Space group		References
	Chemical	Magnetic	
Fe	$Im\bar{3}m$	$I4/mmm'm'$	Sirotn and Shaskalskaya [4]
MnO	$Fm\bar{3}m$	$C_{2c} 2/m$	Prandl [2]
MnF ₂	$P4/mmm1'$	$P4'_2/mmm'$	Sirotn and Shaskalskaya [4]
FeSO ₄	$Cmcm$	$C_p m'cm'$	Prandl [2]

Table 10.4 Results of conventional and magnetic crystallography

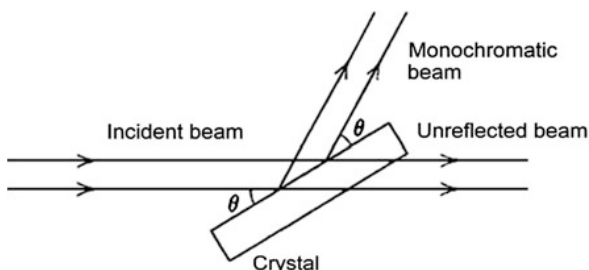
Feature	Conventional crystallography	Magnetic crystallography
Primitive lattices	7	7
Bravais lattices	14	36
Point groups	32	90
Space groups	230	1421

10.3 Neutron Diffraction

10.3.1 General

The development of X-ray diffraction method paved the way for unraveling the structure of crystals. X-rays are scattered by atomic electrons. Therefore, what X-ray diffraction finally reveals is the electron density distribution and the atomic positions. This information is called the crystal structure or, sometimes, the chemical crystal structure.

De Broglie introduced the concept of matter waves having a wavelength $\lambda = h/mv$ where h is Planck's constant, m the particle mass and v the particle velocity. This idea was confirmed by the discovery of the diffraction of electrons by crystals. Here again the electron waves are scattered by the atomic electrons. Consequently, electron diffraction, like X-ray diffraction, yields electron density, atomic positions and crystal structure.

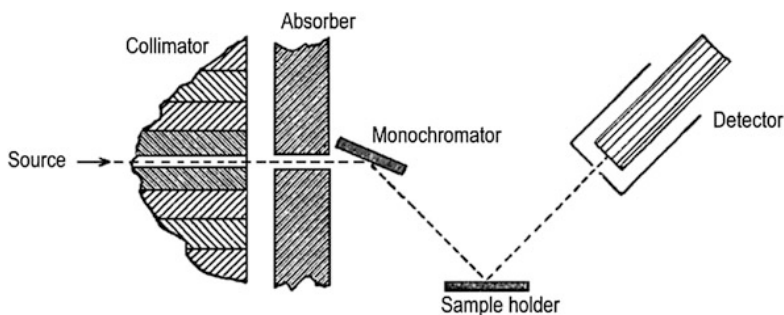
Fig. 10.8 Basic principle of neutron diffraction

In course of time, the diffraction of neutrons by crystals was also discovered. Unlike X-rays, the neutron waves are scattered by the atomic nuclei. This scattering takes place by two mechanisms: (i) nuclear scattering which is due to the collisions between the neutrons and the nuclei and (ii) magnetic scattering which is due to interaction of the magnetic moment of the neutron with that of the atom. The former provides information about the nuclear structure which is essentially the same as the chemical structure provided by X-rays. The latter on the other hand provides information about the magnetic structure which is the symmetric arrangement of ordered spins.

The basic principle of diffraction of neutrons is the same as that of diffraction of X-rays. This is shown in Fig. 10.8. A beam of neutrons is incident on a crystal. Depending on the wavelength λ of the neutron waves and the interplanar spacing d of the crystal, the neutron waves are diffracted at an angle θ such that Bragg's law $n\lambda = 2d \sin \theta$ is satisfied; n is an integer.

10.3.2 Neutron Diffractometer

The basic components of the experimental arrangement for recording a neutron diffraction pattern are shown in Fig. 10.9. They are (a) the source, (b) the collimator, (c) the absorber, (d) the monochromator, (e) the sample holder and (f) the detector. We shall now discuss the details of these components.

**Fig. 10.9** Basic components of a neutron diffractometer

- (a) In pioneering work, the source of neutrons was a Ra-Be source. However, it is now completely superseded by a nuclear reactor.
- (b) Most materials are almost completely transparent to neutrons because of the uncharged nature of the neutrons. Lead, however, is a comparably good absorber. Hence collimation is achieved by long thick tubes or blocks of lead with a channel for the neutrons.
- (c) The wavelength of neutrons depends on the velocity through the De Broglie relation $\lambda = h/mv$. The neutrons produced in a reactor have a velocity distribution. In the reactor, the neutrons undergo collisions with the atoms of what is called a moderator in which their velocity gets reduced. Finally, they achieve equilibrium appropriate to the temperature of the moderator. This determines the mean velocity of the neutrons. For a temperature of ~ 320 K of the moderator, the mean velocity is such that the wavelength is ~ 1.4 Å; this velocity is suitable for diffraction of neutrons. Such neutrons are called thermal neutrons. But even thermal neutrons have a velocity spread. Hence, in neutron diffraction, monochromatization means selecting neutrons with some desired velocity.

In early experiments, a device called chopper (Fig. 10.10a) was used. It consisted of two discs made of an absorber each having a window; the two windows are in a line so that neutrons entering through one window can pass through the other. The discs are mounted on a shaft which can be rotated. Neutrons entering through the first window can pass through the second window only if their velocity is equal to l/t where l is the distance between the two discs and t the period of rotation of the chopper. Neutrons with other velocities are absorbed by the second disc.

However, nowadays, the phenomenon of diffraction of neutrons is itself used for monochromatization. If a crystal with its d value predetermined by X-rays is used, it will diffract only the wavelength that satisfies Bragg's law (Fig. 10.10b). Single crystals of Cu, Pb, Zn, Ge and pyrolytic graphite are employed for the purpose [7].

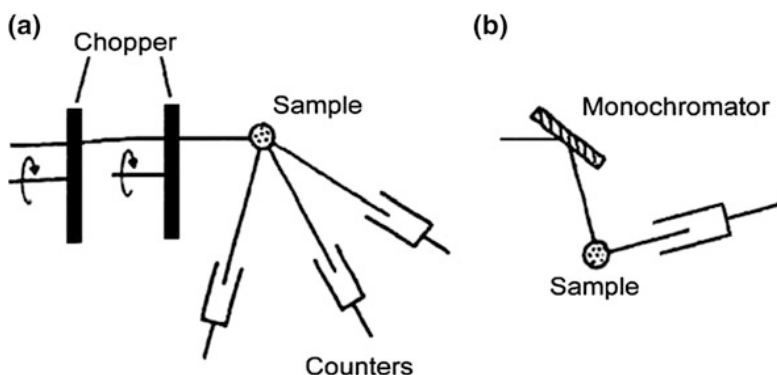


Fig. 10.10 Types of monochromators; **a** Chopper, **b** Crystal diffraction

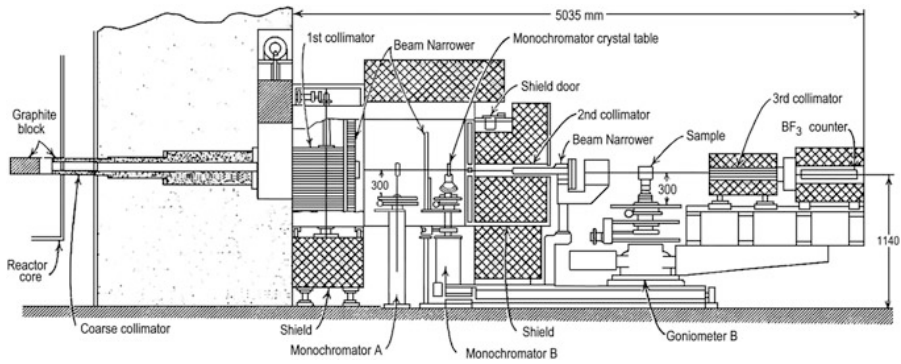


Fig. 10.11 Details of a neutron diffractometer

- (d) If the material is in the form of powder, the sample is in the form of a vanadium capsule containing the powder or a compressed cylinder. In view of the low absorption by the diffracting material, the size of the sample is large compared with that in X-ray diffraction. Typical size is 2.5 cm in diameter and 5 cm in length. Single crystal samples are mounted on a goniometer which has provisions for orienting the crystal.
- (e) The detector is generally a BF_3 counter mounted on a rotatable table. The actual equipment is much more elaborate. A lateral view is shown in Fig. 10.11. The figure is self-explanatory. Multiple collimation and multiple monochromatization is employed. Heavy shielding (cross-hatched regions in the figure) is used to avoid multiple scattering of neutrons in space. Note also the dimensions; neutron diffractometer occupies quite some space.

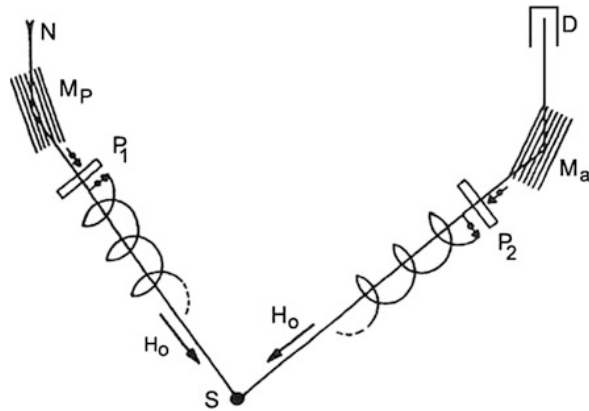
10.3.3 Polarized Neutrons

For some studies, a beam of polarized neutrons is used. The principle of a polarized neutron spectrometer is shown in Fig. 10.12. The incident beam of unpolarized neutrons is diffracted by a crystal which is magnetized in the vertical direction. The diffracted beam is partially polarized. If the neutrons are polarized in the same direction as the magnetized crystal (called the polarizer) the differential cross-section $\left(\frac{d\sigma}{\Omega}\right)_1$ is

$$\left(\frac{d\sigma}{\Omega}\right)_1 = (C - D)^2 \quad (10.1)$$

where C and D are the amplitudes of the nuclear scattering and magnetic scattering. On the other hand, if the neutron polarization is antiparallel to the magnetization of the polarizer, the differential cross-section $\left(\frac{d\sigma}{\Omega}\right)_2$ is

Fig. 10.12 Polarized neutron diffractometer. N unpolarized incident beam; M_p , M_a polarizer and analyzer, P_1 , P_2 $\pi/2$ spin-turn coils, S Sample, H_0 guide field, D detector



$$\left(\frac{d\sigma}{\Omega}\right)_2 = (C + D)^2 \quad (10.2)$$

If the polarizer crystal is chosen such that $C = D$, $\left(\frac{d\sigma}{\Omega}\right)_1$ vanishes and we get only a negatively polarized neutron beam. It is found that this condition is satisfied by the (111) plane of magnetite (Fe_2O_3) and the (200) plane of Fe-Co crystal.

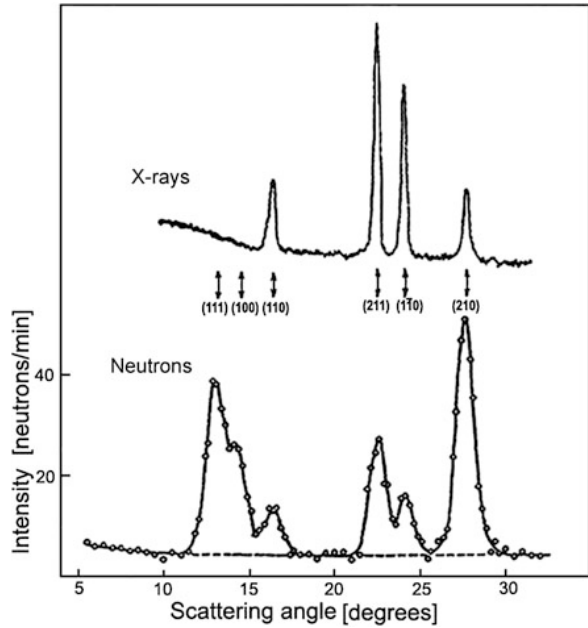
The diffracted beam then passes through a magnetic field parallel to the polarization of the neutrons. To reverse the polarization of the beam, the magnetic field of a r.f. field is employed which 'flips' the neutron. By using both the beams, nuclear scattering is almost completely eliminated and only magnetic scattering prevails.

The advantages in using polarized neutrons and also their applications are discussed by Hayter [8] and Satya Murthy and Paranjape [9].

10.3.4 Analysis of Neutron Diffraction Data

X-ray and neutron diffraction patterns for the powder samples of a magnetic material are shown in Fig. 10.13. Some reflections in the X-ray pattern are repeated in the neutron diffraction pattern. These represent the chemical crystal structure of the sample. The reflections in the X-ray pattern are caused by the scattering of X-rays by electrons and those in the neutron diffraction pattern by nuclear scattering. The extra lines occurring in the neutron diffraction pattern are due to the magnetic scattering by atoms with non-zero spins which are in an ordered state (the magnetic structure). In the discussion that follows, our emphasis will be on these extra reflections.

Fig. 10.13 X-ray and neutron diffraction patterns of a magnetic material



10.3.4.1 Lattice Constants

From the Bragg angles (θ) and the Miller indices (h, k, l) of the reflections, the lattice constants can be calculated from the relations:

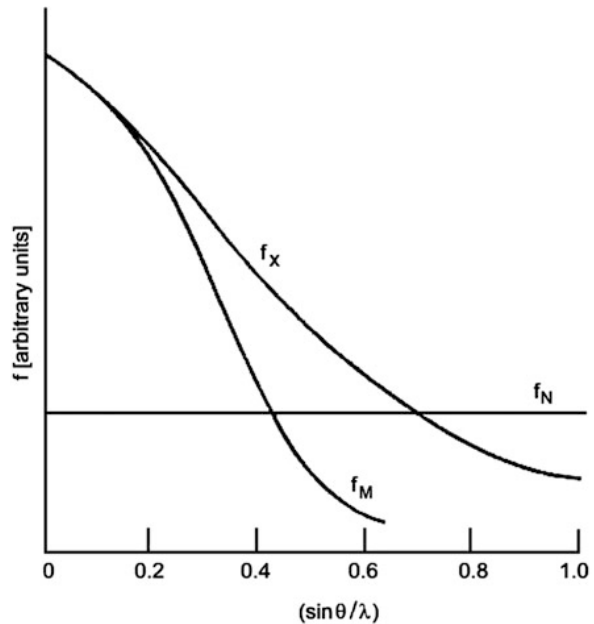
$$\left. \begin{aligned} \text{For cubic crystals : } \sin^2 \theta &= \left(\frac{\lambda^2}{4a^2} \right) (h^2 + k^2 + l^2). \\ \text{For tetragonal crystals : } \sin^2 \theta &= \left(\frac{\lambda^2}{4a^2} \right) (h^2 + k^2) + \left(\frac{\lambda^2}{4c^2} \right) l^2. \\ \text{For tetragonal crystals : } \sin^2 \theta &= \left(\frac{\lambda^2}{3a^2} \right) (h^2 + hk + k^2) + \left(\frac{\lambda^2}{4c^2} \right) l^2. \end{aligned} \right\} \quad (10.3)$$

Here a is the lattice constant for a cubic crystal and a and c are the lattice constants for tetragonal and hexagonal crystals. Similar relations for other crystal systems can be found in text books on X-ray diffraction.

10.3.4.2 Intensities

Intensities of the diffraction beams are the most important input for the determination of the structure. In order to understand the mechanism of neutron diffraction, it is useful to start with the mechanism of X-ray diffraction. X-rays are scattered by the electrons in an atom. The resultant of the amplitudes of waves scattered by all

Fig. 10.14 X-ray and nuclear and magnetic form factors of an atom as a function of $(\sin \theta/\lambda)$



the electrons in the atom is called the atomic scattering factor or the X-ray form factor f_X . In a similar way the neutrons are scattered by the nucleus. The form factor due to neutron-nucleus interaction is called the nuclear form factor and is denoted by f_N . The neutron has spin and an associated magnetic moment. Atoms with unpaired electrons (non-zero spin) also have a magnetic moment. The interaction between neutron magnetic moment and the atomic magnetic moment results in scattering which is called magnetic scattering. The form factor for this kind of scattering is denoted by f_M . Values of f_X , f_N and f_M are plotted against $(\sin \theta/\lambda)$ in Fig. 10.14; here θ is the Bragg angle. It may be noted that f_N is constant. Both f_X and f_M decrease with $(\sin \theta/\lambda)$; the decrease with $(\sin \theta/\lambda)$ is faster for f_M .

We shall now consider the concept of 'structure factor'. This is the ratio of the resultant amplitude of waves scattered by all atoms in the unit cell to the amplitude of the wave scattered by a single electron. For X-rays, the structure factor for the plane with Miller indices hkl is

$$F_X(hkl) = \sum_j f_{Xj} \exp i\phi_j. \quad (10.4)$$

Here ϕ_j is the phase; it is given by

$$\phi_j = 2\pi(hu_j + kv_j + lw_j) \quad (10.5)$$

where u_j , v_j , w_j are the coordinates of the j th atom in the unit cell. The coordinate is the actual Cartesian coordinate expressed as a fraction of the lattice constant.

Similarly, when neutron waves are diffracted, the nuclear structure factor has the form

$$F_N(hkl) = \sum_j f_{Nj} \exp 2\pi i(hu_j + kv_j + lw_j) \quad (10.6)$$

where f_{Nj} is the nuclear form factor for the j th nucleus. If the unit cell contains some atoms with non-zero spin and if the spins are in an ordered state, the neutrons are magnetically scattered and the magnetic structure factor for these waves is

$$F_M(hkl) = \sum_{j'} f_{Mj'} \exp 2\pi i(hu_{j'} + kv_{j'} + lw_{j'}) \quad (10.7)$$

where $u_{j'}$, $v_{j'}$ and $w_{j'}$ are the coordinates of the j' th atom with non-zero spin. Note that in (10.7), $f_{Mj'}$ can take positive or negative values depending on whether the spin of that atom is in the up-state or down-state.

Finally, the integrated intensity I_{hkl} of a Bragg reflection of scattered neutrons is given by

$$I_{hkl} = kI_0 A_{hkl} J_{hkl} e^{-2W} |F_{hkl}|^2 / \sin 2\theta \quad (10.8)$$

in which kI_0 is an instrumental constant, A_{hkl} the absorption correction, J_{hkl} the multiplicity, e^{-2W} the Debye-Waller correction for thermal motion and F_{hkl} is the structure factor [10]. In structure determination $|F_{hkl}|^2$ is taken as the intensity.

In neutron diffraction, if both nuclear scattering and magnetic scattering contribute to a reflection, the structure factor squared is given [11] as

$$|F_{hkl}|^2 = |F_N(hkl)|^2 + q^2 |F_M(hkl)|^2 \quad (10.9)$$

Here $q^2 = \sin^2 \theta$ where θ is the angle between the normal to the reflecting plane and the direction of magnetization of the sample. Thus $0 \leq q^2 \leq 1$. Special values of q^2 are 1 when the directions are perpendicular and 0 when they are parallel; $q^2 = 2/3$ for an unmagnetized (paramagnetic) sample. In the literature on neutron diffraction, $|F_N(hkl)|$ and $|F_M(hkl)|$ are denoted by C and D , a practice introduced by Halpern and Johnson [11].

In determining magnetic structures, the intensities are used in different ways; we shall discuss some of these.

- (i) In some structures, select reflections have zero intensity in the non-magnetic state and non-zero intensity in the magnetic state. These are extra reflections which appear in the neutron diffraction pattern of the magnetic sample. The mere presence or absence of these reflections throws light on the type of magnetic order.

As an example, let us consider a bcc lattice with identical atoms at the corner (0, 0, 0) and at the body-centre ($\frac{1}{2}, \frac{1}{2}, \frac{1}{2}$). In the non-magnetic state, when only nuclear scattering contributes, the structure factor is

$$\begin{aligned} F_N(hkl) &= f_N [\exp 2\pi i(h \cdot 0 + k \cdot 0 + l \cdot 0) + \exp 2\pi i(h \cdot \frac{1}{2} + k \cdot \frac{1}{2} + l \cdot \frac{1}{2})] \\ &= f_N [1 + \exp \pi i(h + k + l)] \end{aligned} \quad (10.10)$$

Thus $F_N = 0$ for $(h + k + l)$ odd and $F_N \neq 0$ for $(h + k + l)$ even. Hence, the (100) reflection will be absent but (200) will be present. If the substance is in the ferromagnetic state, then the structure factor is similar with f_M substituting f_N i.e.

$$F_M(hkl) = f_M [1 + \exp \pi i(h + k + l)] \quad (10.11)$$

Now again, the (100) reflection will be absent and the (200) reflection will be present. On the other hand, if the substance is antiferromagnetic, the form factor of the waves from the corner atom and the body-centre will be different (say f_{M1} and f_{M2}) and the structure factor is

$$F_M(hkl) = f_{M1} + f_{M2} \exp \pi i(h + k + l). \quad (10.12)$$

It can be seen that the structure factor and hence the intensity for (100) is no longer zero. The presence of the (100) reflection leads to the conclusion that the substance is antiferromagnetic.

- (ii) If a clear indication of magnetic structure is not available from intensities of individual reflections, then, as done in X-ray structure determination, a spin arrangement is assumed and theoretical intensities are calculated from (10.6)–(10.9). The calculated intensities are compared with observed intensities. If there is over-all agreement, the assumed structure is taken as correct. Otherwise another structure is assumed and the process is repeated until we reach the correct structure.
- (iii) In X-ray diffraction, a Fourier synthesis of the intensities leads to an electron density distribution map from which atomic positions can be located. From the Fourier synthesis of the intensities of a magnetic crystal, information can be obtained about the spin orientations.

10.4 Examples of Magnetic Structures

10.4.1 General

Experimental results on the temperature variation of magnetic susceptibility clearly distinguish the various magnetic types. Some information on magnetic structures is provided by properties like magnetic anisotropy and magnetostriction. X-ray

diffraction provides information regarding the basic chemical structure. Clinching evidence about the magnetic structure i.e. the symmetric orderly alignment of spins is provided by neutron diffraction. The principle and practice of neutron diffraction has been discussed in the preceding section. Magnetic structures of a large number of crystals have been reported; we shall discuss a few of them.

10.4.2 Ferromagnetic Structures

The magnetic structures of iron and cobalt have been studied by Shull et al. [12]. Fe has the bcc structure (Fig. 10.15a). It has a Curie temperature of 1,043 K and so it is ferromagnetic at room temperature. The neutron diffraction pattern of Fe powder at room temperature is shown in Fig. 10.16. The reflections can be indexed on the basis of the bcc structure assuming the same lattice constant as derived from the X-ray diffraction pattern. There are no extra reflections. Assuming that the reflections are due to nuclear scattering, their intensities can be calculated from the theoretical nuclear form factors. The observed intensities are systematically larger than the calculated intensities. Obviously there is a strong magnetic contribution. From the difference between the observed intensities and the calculated intensities, the magnetic form factors can be estimated. These form factors are shown in Fig. 10.17 as a function of $(\sin \theta/\lambda)$ along with the nuclear form factor. The experimental magnetic form factor shows the same variation with $(\sin \theta/\lambda)$ as is observed in the theoretical magnetic form factor (Fig. 10.14), thus confirming that Fe is in the ferromagnetic state. The spin orientation in Fe is shown in Fig. 10.15b. It has been mentioned that from a Fourier synthesis of the intensities, the spin density distribution map (sddm) can be constructed. The sddm for Fe is shown in Fig. 10.18. It is seen that the spin density spreads more along [100] direction than along the [110] direction [13].

The results on Co are similar to those on Fe.

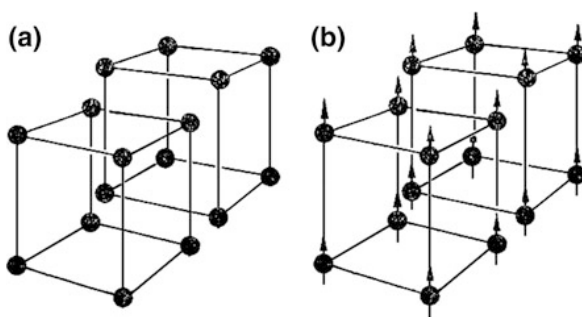


Fig. 10.15 Body-centred cubic structure of iron; **a** In paramagnetic state without ordered spins, **b** In ferromagnetic state with parallel ordering of spins

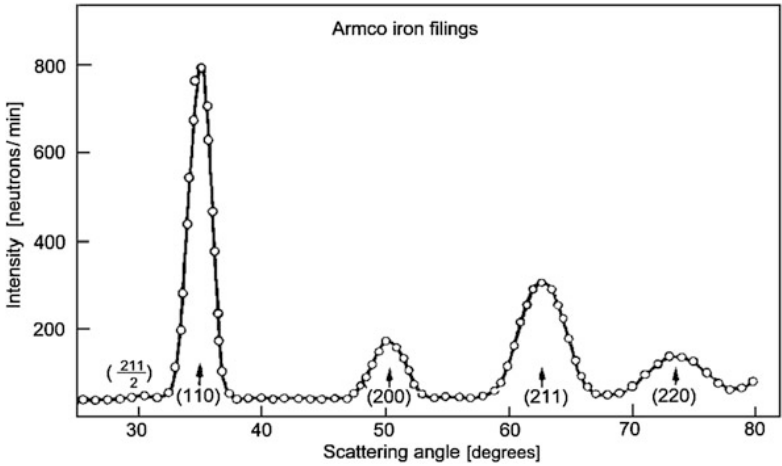


Fig. 10.16 Neutron diffraction pattern of ferromagnetic iron

Fig. 10.17 Magnetic scattering form factor (expressed as the differential cross-section) of iron as a function of $(\sin \theta/\lambda)$; *dashed curve* nuclear scattering form factor

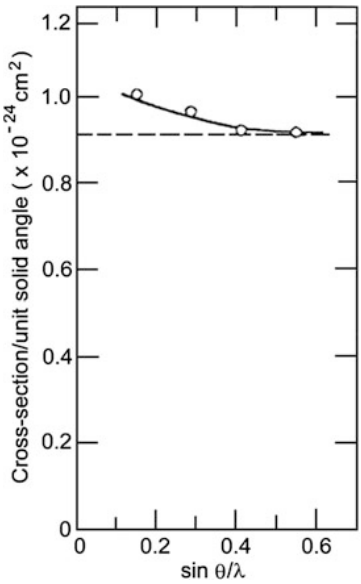
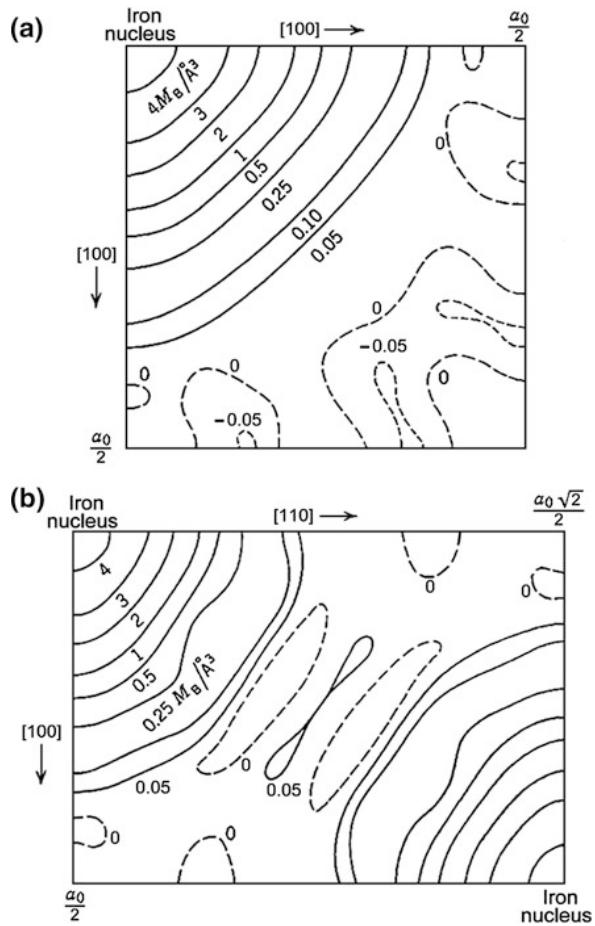


Fig. 10.18 Spin density distribution of iron **a** On (001) plane, **b** On (110) plane



10.4.3 Antiferromagnetic Structures

10.4.3.1 Transition Elements

Like Fe and Co, the transition metals chromium and α -manganese also have bcc structure (Fig. 10.19a). Magnetic susceptibility measurements show that they are antiferromagnetic with $T_N = 475$ and 100 K respectively.

Shull and Wilkinson [14] carried out neutron diffraction studies on Cr. The neutron diffraction pattern recorded at 20 K is shown in Fig. 10.20. The (110), (200), (211) reflections are due to nuclear scattering. It was pointed out in Sect. 10.3 that in antiferromagnetic bcc lattices, the (100) reflection should appear. This reflection is seen in the pattern. Further the intensity of the (100) reflection (Fig. 10.21) decreases with increasing temperature, finally approaching zero at $T \approx 475$ K. Also, the shape of the temperature versus intensity curve resembles a

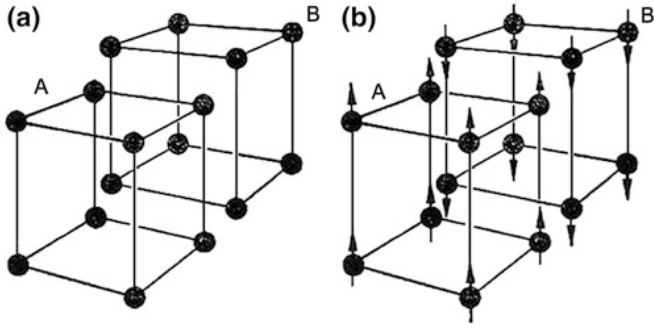


Fig. 10.19 Body-centred cubic structure of Cr and α -Mn; **a** In paramagnetic state without ordered spins, **b** In ferromagnetic state with antiparallel ordering of spins

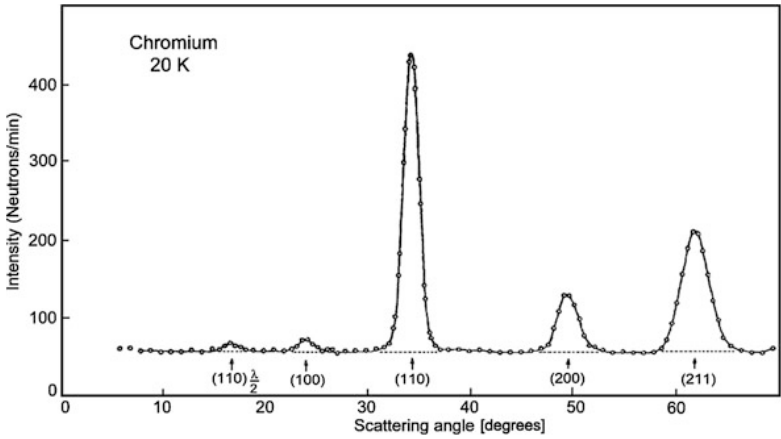


Fig. 10.20 Neutron diffraction pattern for Cr taken at 20 K; note the presence of the (100) magnetic reflection

Brillouin function. These features clearly indicate that the (100) reflection is due to antiferromagnetism in Cr. The spin arrangement is shown in Fig. 10.19b.

Shull and Wilkinson [14] also studied α -Mn. The neutron diffraction patterns recorded at 295 and 20 K are shown in Fig. 10.22. The reflections in the 295 K pattern are due to nuclear scattering. The pattern at 20 K also shows these reflections along with extra reflections due to antiferromagnetism. The intensities of the extra reflections decrease with increasing temperature and approach zero value at $T \approx 100$ K showing that they are related to antiferromagnetism. However, the (100) reflection which is seen in the Cr pattern in Fig. 10.20 does not show up in the 20 K pattern (Fig. 10.22). This suggests that the antiferromagnetism in Mn is weaker than that in Cr.

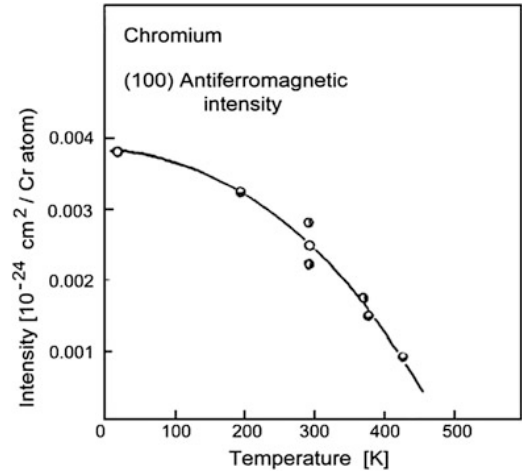


Fig. 10.21 Temperature variation of the intensity of the (100) magnetic reflection

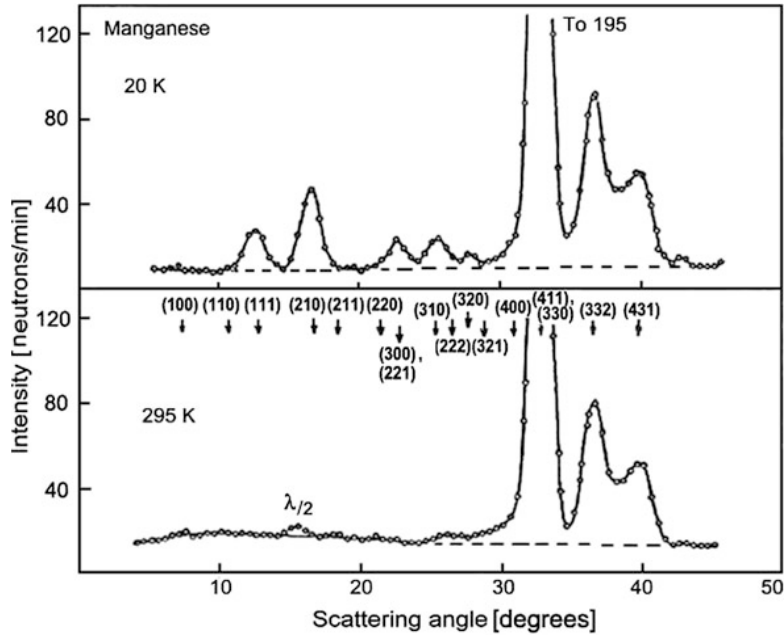


Fig. 10.22 Neutron diffraction patterns of α -Mn at 20 and 295 K

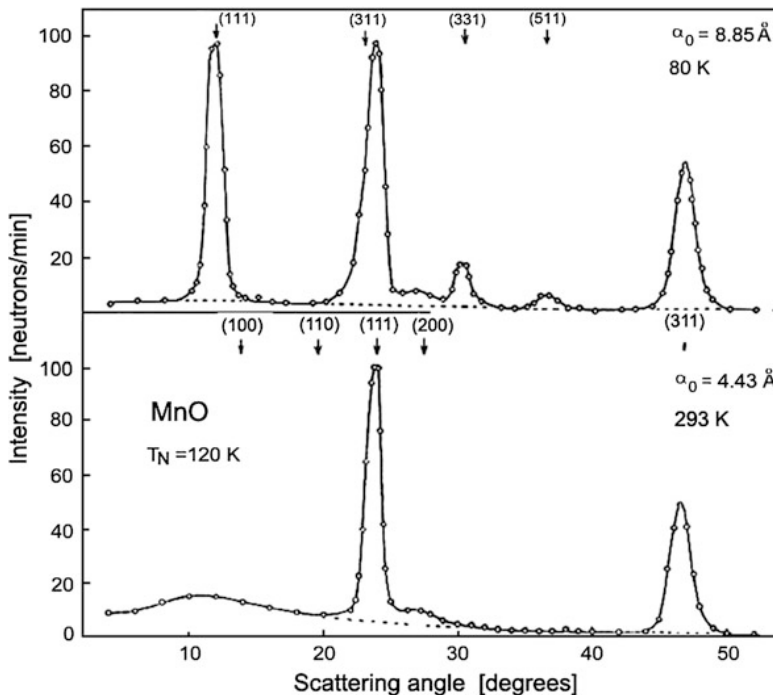


Fig. 10.23 Neutron diffraction patterns for MnO above and below the Neel temperature

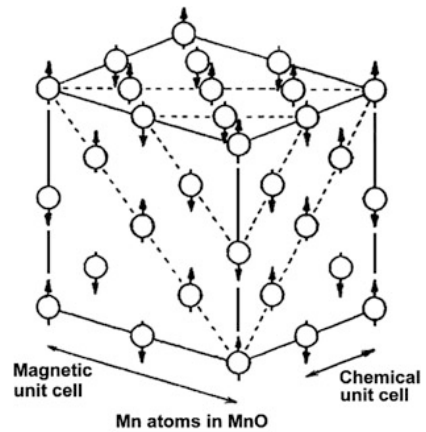
10.4.3.2 Transition Metal Oxides

Manganese oxide (MnO) is cubic with NaCl structure and a lattice constant $a = 4.45 \text{ \AA}$. It is antiferromagnetic with $T_N = 120 \text{ K}$. MnO was the first ever magnetic structure to be explored with neutron diffraction.

The neutron diffraction patterns of MnO recorded by Shull and Smart [15] at 293 K (i.e. above T_N) and at 80 K (i.e. below T_N) are shown in Fig. 10.23. The room temperature pattern is that expected from a crystal with NaCl structure having $a = 4.45 \text{ \AA}$. The X-ray pattern of MnO (not shown here) has strong hkl -all-even reflections and weak hkl -all-odd reflections. In contrast, the neutron diffraction pattern has weak hkl -all-even reflections and strong hkl -all-odd reflections. This is because the signs of nuclear scattering amplitudes are of opposite sign for the Mn and O ions.

Both the neutron diffraction patterns shown in Fig. 10.23 conform to the NaCl structure. However, there is a dramatic difference between the two. Whereas the room temperature (RT) pattern can be indexed with a lattice constant of $4.423 \text{ \AA}$, the low temperature (LT) pattern can be indexed with a lattice constant of $8.85 \text{ \AA}$. It is therefore concluded that while the RT pattern is caused by nuclear scattering, the LT pattern is caused by magnetic scattering from the spin moments of Mn ions in antiparallel orientation. The chemical unit cell and magnetic unit cell of MnO are

Fig. 10.24 Magnetic structure of MnO; note the magnetic unit cell edge which is twice the chemical unit cell edge



shown in Fig. 10.24. In the chemical lattice alternate points are occupied by Mn and O ions. In the magnetic lattice alternate points are occupied by spin-up and spin-down ions of Mn. The Mn–Mn distance in the chemical lattice is 4.42 Å. The Mn $\uparrow$ –Mn $\uparrow$ distance in the magnetic lattice is 8.85 Å (the arrow indicates spin-up).

The temperature variation of the intensity of (111) reflection of MnO is shown in Fig. 10.25. The intensity decreases with increasing temperature and tends to zero as the temperature tends to T_N . Further, the temperature versus intensity curve has the shape of a Brillouin function. This shows that the (111) reflection is due to the antiferromagnetic state. It may be mentioned that whereas the space group of MnO is $Fm\bar{3}m$, its magnetic space group is $C_{2c}2/m$.

Shull et al. [16] investigated the orientations of the spins. They assumed three models: (a) magnetic moments aligned along [100] direction as shown in Fig. 10.24 with $q^2 = 2/3$, (b) magnetic moments perpendicular to the (111) sheets with $q^2 = 2/3$ for (111) reflection, 32/33 for (311) and 32/57 for (331) reflections and (c) magnetic moments aligned arbitrarily in the (111) sheets with $q^2 = 1$ for the (111) reflections and various values for the other reflections. The intensities of the magnetic reflections for these three models are given in Table 10.5 along with observed intensities. It is seen that only model (a) gives intensities agreeing with observed intensities.

Shull et al. [14] also studied FeO, NiO and CoO. Their neutron diffraction patterns are similar to that for MnO in the antiferromagnetic state. Thus their magnetic structure is similar to the one shown in Fig. 10.24.

Clifford Shull's experiment on MnO opened up the field of magnetic structures. He was awarded the Nobel Prize for physics in 1994.

Fig. 10.25 Temperature variation of the intensity of the magnetic reflection (111) of MnO

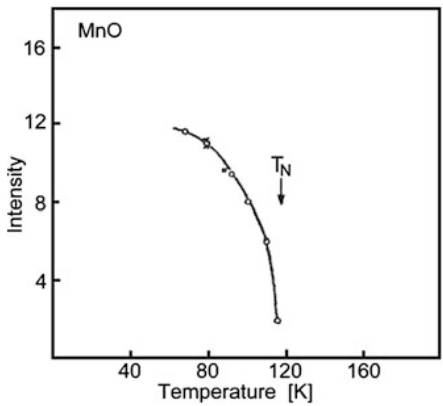


Table 10.5 Observed and calculated neutron diffraction intensities for MnO in antiferromagnetic state

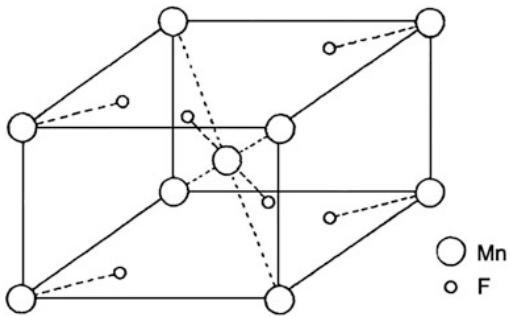
	Calculated for various models			Observed
	(a)	(b)	(c)	
(111)	1,038	0	1,560	1,072
(311)	460	675	—	308
(331)	129	109	—	132
(511), (333)	54	24	—	70

10.4.3.3 MnF₂ and Isomorphous Crystals

MnF₂ has the tetragonal rutile structure shown in Fig. 10.26. Magnetic susceptibility measurements show that MnF₂ is antiferromagnetic with $T_N = 75$ K.

The magnetic structure of MnF₂ was studied by Erickson [17]. The neutron diffraction patterns obtained by him at RT and at 23 K (LT) are shown in Fig. 10.27. The pattern at RT consists of reflections due to nuclear scattering by the chemical unit cell. The pattern at LT shows the same reflections (as in the RT

Fig. 10.26 Crystal structure of MnF₂



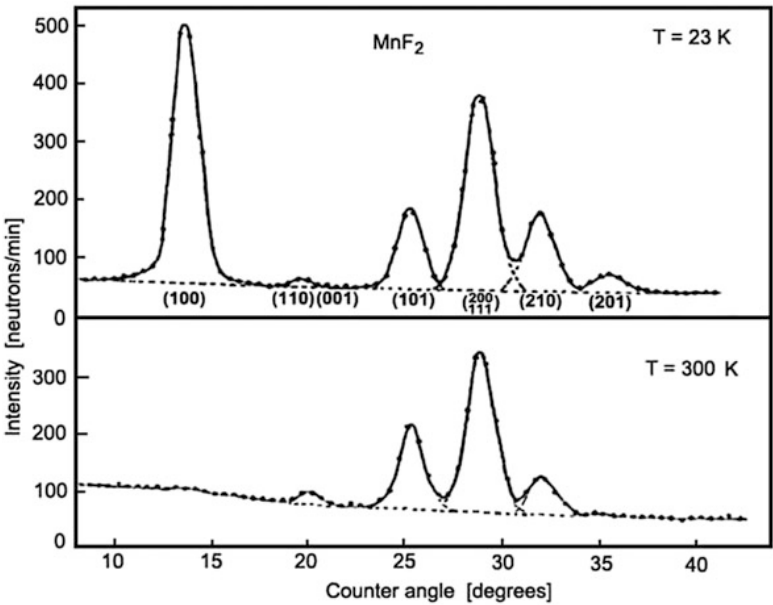
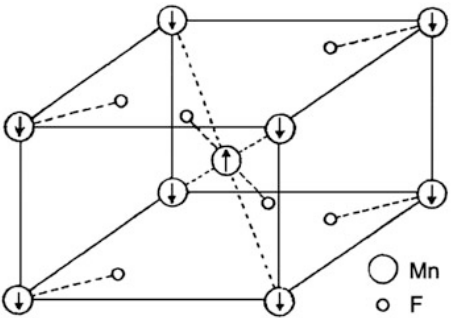


Fig. 10.27 Neutron diffraction patterns for MnF_2 in the paramagnetic state (300 K) and in the antiferromagnetic state (23 K)

pattern) and, in addition, some new reflections. The presence of the former shows that the chemical unit cell at LT is the same as that at RT. The presence of the latter reflections has to be attributed to the magnetic superlattice. If the magnetic lattice is the same as the chemical lattice, the new reflections can be indexed as (100), (111), (210) and (201). The chemical unit cell has two Mn^{2+} ions at (000) and $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$. The spin moments of these two ions are antiparallel. The absence of the (001) reflection indicates that the spin moments are aligned along the c -axis.

The magnetic structure of MnF_2 is shown in Fig. 10.28. The environment of each Mn^{2+} ion can be described as two parallel neighbours at 3.31 Å along the c -axis, eight antiparallel neighbours at 3.81 Å, four parallel neighbours at 4.87 Å and

Fig. 10.28 Magnetic structure of MnF_2



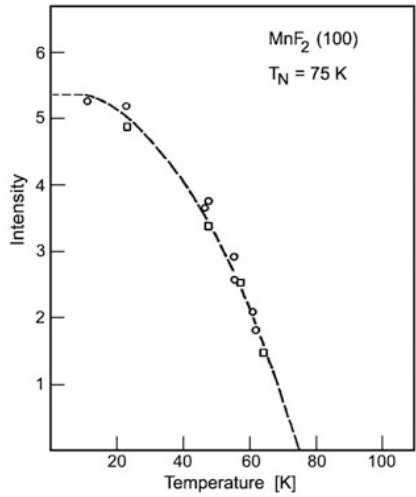


Fig. 10.29 Temperature variation of the (100) magnetic reflection of MnF_2

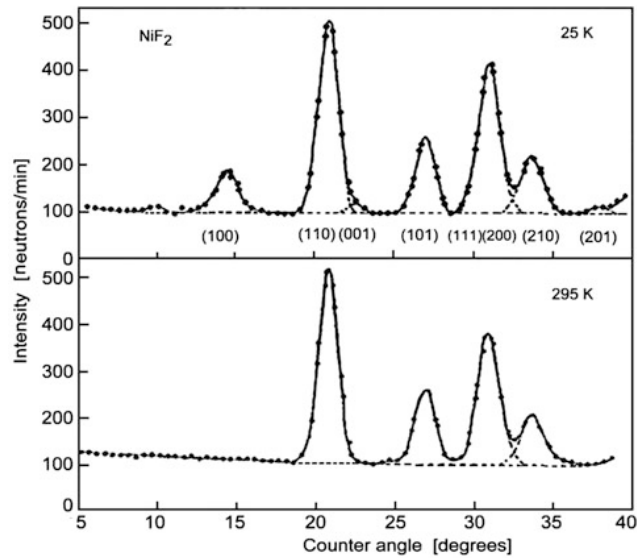


Fig. 10.30 Neutron diffraction patterns for NiF_2 at 295 and 25 K; note presence of magnetic reflection (001) in the low temperature pattern

eight antiparallel neighbours at 6.03 Å. Thus, the spin ordering in MnF_2 is antiferromagnetic.

The temperature variation of the intensity of the (100) reflection is shown in Fig. 10.29. It decreases with increasing temperature and tends to zero at T_N . This confirms that the origin of the extra lines is in the antiferromagnetism of MnF_2 .

Isomorphous compounds CaF_2 and FeF_2 have been studied by Erickson [17] and have shown a similar magnetic structure as MnF_2 . In NiF_2 , which is otherwise isomorphous to MnF_2 , the (001) reflection (absent in MnF_2) appears with an unmistakable intensity (Fig. 10.30). This indicates that the spins are not exactly parallel to the c -direction but make an angle of 10° to the axis.

10.4.3.4 Fe_2O_3 (Hematite)

$\alpha\text{-Fe}_2\text{O}_3$ (Hematite) is trigonal with $a = 5.42 \text{ \AA}$ and $\alpha = 55^\circ 17'$. There are two molecules in the unit cell (Fig. 10.31). Note the Fe positions; these Fe ions lie in sheets in the (111) plane (or normal to the [111] direction). Magnetic susceptibility measurements by Morin [18] show that $\alpha\text{-Fe}_2\text{O}_3$ is antiferromagnetic with $T_N = 950 \text{ K}$.

Shull et al. [16] recorded X-ray diffraction pattern and neutron diffraction pattern of Fe_2O_3 at room temperature. These patterns are shown in Fig. 10.32. The reflections present in the X-ray pattern are also present in the neutron diffraction pattern indicating that these reflections are due to nuclear scattering. In addition, the neutron diffraction pattern contains lines which can be indexed as (111) and (100); since these are not present in the X-ray pattern, these must be due to magnetic scattering. In fact, as we shall see, much information can be extracted from these reflections.

To decide the magnetic structure of $\alpha\text{-Fe}_2\text{O}_3$, Shull et al. [16] calculated the intensities of the (111) and (100) reflections from different models. Designating the four Fe ions as A, B, C, D and a spin-up by + and a spin-down by –, the three assumed models are: (a) + – – +, (b) + + – – and (c) + – + –. The calculated structure factors for the (111) and (100) reflections are zero for model (c); hence it is ruled out. The calculated intensities will depend not only on models (a) and (b) but also on the assumed orientations of the spins. Here, again, there are three possibilities. These are:

- I. The spin moments are directed along the unit cell edges.
- II. The moments are directed along the space diagonal of the unit cell and are thus perpendicular to the (111) Fe sheets.
- III. The moments are in the (111) sheets and directed towards one of the three nearest neighbours in the sheet.

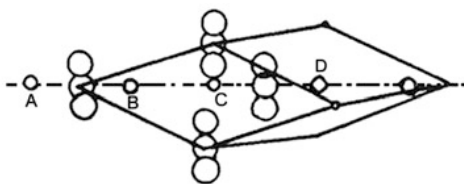


Fig. 10.31 Chemical structure of $\alpha\text{-Fe}_2\text{O}_3$; *small circles* represent Fe ions while *large circles* represent oxygen ions

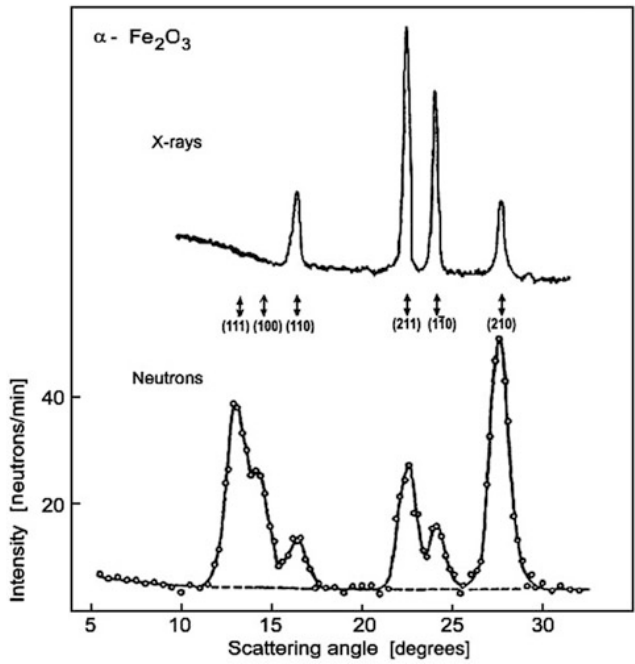


Fig. 10.32 X-ray and neutron diffraction patterns of $\alpha\text{-Fe}_2\text{O}_3$; innermost two reflections in the neutron diffraction pattern are antiferromagnetic in origin and are absent in the X-ray pattern

Table 10.6 Comparison between observed and calculated F^2 values for various magnetic structure models of $\alpha\text{-Fe}_2\text{O}_3$

$F^2 \rightarrow$	Calculated						Observed	
	Model (a)			Model (b)				
$hkl \downarrow$	(I)	(II)	(III)	(I)	(II)	(III)	300 K	80 K
(111)	1.25	0	4.3	0.23	0	0.81	4.9	<0.05
(100)	1.40	1.59	0.96	2.32	2.64	1.59	0.91	1.37

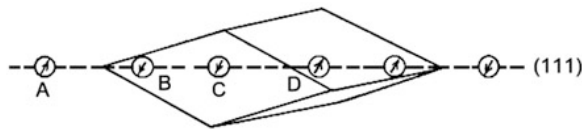


Fig. 10.33 Magnetic structure of $\alpha\text{-Fe}_2\text{O}_3$; only the iron ions are shown

Intensities were calculated for the two reflections at 300 and 80 K. They are given in Table 10.6 along with observed intensities. It is seen that RT data favours model (a) with spin orientation III whereas LT data favours model (a) with spin orientation II (Fig. 10.33).

10.4.3.5 LiFePO₄

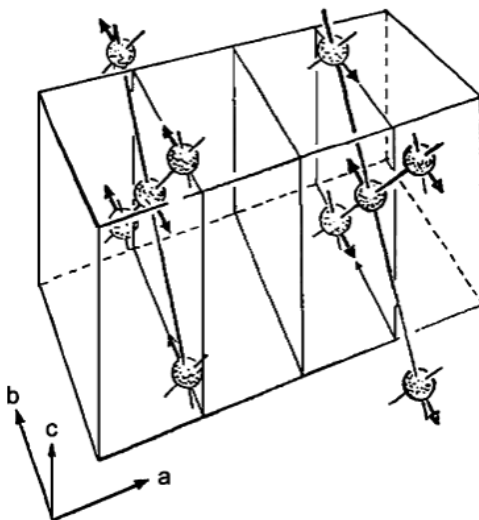
Magnetic susceptibility measurements, X-ray structure determination and magnetic structure determination of LiFePO₄ was carried out by Santoro and Newnham [19].

Magnetic susceptibility measurements made over the range 20–300 K showed that LiFePO₄ is antiferromagnetic with $T_N \approx 50$ K. X-ray powder diffraction studies at RT revealed that LiFePO₄ is orthorhombic with $a = 10.31$ Å, $b = 5.997$ Å and $c = 4.686$ Å; there are four molecules in the unit cell. Intensity analysis showed that LiFePO₄ has the olivine structure with space group $Pnma$.

Neutron diffraction patterns were recorded at 300 K (i.e. above T_N) and at 9 K (i.e. below T_N). The patterns at LT had reflections common with the X-ray pattern. The intensities calculated with X-ray-determined parameters agreed with observed intensities showing that there is no change in the chemical unit cell.

In the olivine structure, the Fe²⁺ ions have coordinates (0.28, $\frac{1}{4}$, 0.98), (0.72, $\frac{3}{4}$, 0.02), (0.22, $\frac{3}{4}$, 0.48) and (0.78, $\frac{1}{4}$, 0.52). Antiferromagnetic spin orientations $+- -+$ are assigned to the Fe ions. If the spins are assumed to be collinear with a or c , a large intensity is predicted for the (010) reflection. But this reflection is not observed in the recorded pattern. It was concluded that the spins are collinear with the b direction. With this assignment of spin, there is good agreement between calculated and observed intensities particularly for the strong (101) and (201) reflections. While the space group of the chemical structure is $Pnma$, the space group of the magnetic structure is $Pnma'$. The magnetic structure is shown in Fig. 10.34.

Fig. 10.34 Magnetic structure of LiFePO₄; only the iron ions are shown



10.4.3.6 CrSe, An Umbrella Structure

The magnetic structures that have been discussed so far were all collinear i.e. half the spins were oriented in one direction and the other half in the opposite direction. We shall now consider the magnetic structure of CrSe which is not collinear but has a novel spin ordering.

CrSe has the hexagonal NiAs structure (Fig. 10.35). The unit cell dimensions measured with X-rays are $a_0 = 3.674 \text{ \AA}$ and $c_0 = 6.001 \text{ \AA}$. The space group is $P6_3/mmc$. From susceptibility, it is concluded that CrSe is antiferromagnetic with $T_N = 300 \text{ K}$.

The magnetic structure of CrSe was investigated by Corliss et al. [20]. They recorded the neutron diffractions patterns at RT and 77 K (Fig. 10.36). The RT pattern could be indexed on the basis of the NiAs structure using nuclear scattering amplitudes. The 77 K pattern could be indexed only on the basis of a unit cell with

Fig. 10.35 Chemical structure of CrSe

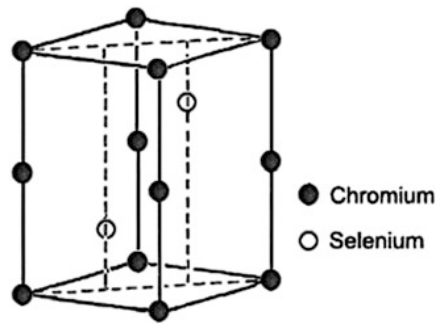
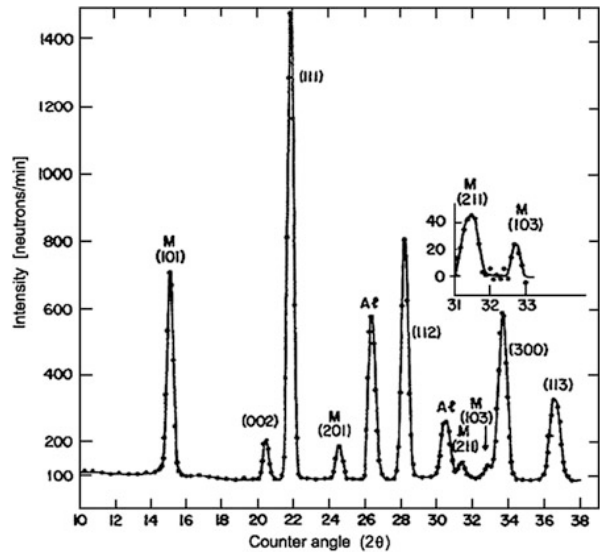


Fig. 10.36 Neutron diffraction pattern for CrSe at 77 K. The indexing is based on an enlarged cell. Magnetic reflections are indicated by symbol *M*. Inset shows enlargement of (211) and (103) reflections



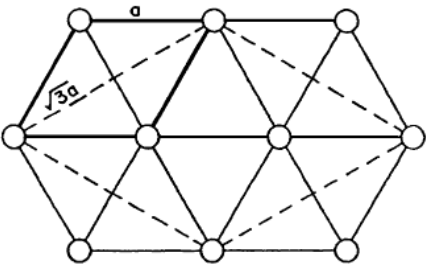
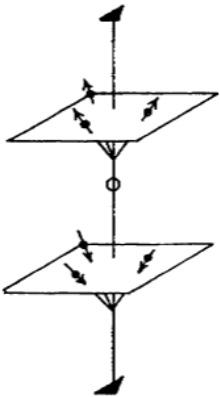


Fig. 10.37 Projection of unit cell of CrSe on basal plane. Chemical cell is shown by *dark lines* and magnetic cell by *dashed lines*

Table 10.7 Comparison of observed and calculated intensities for CrSe

<i>hkl</i>	<i>I</i> _{calc.}		<i>I</i> _{obs.}
	Collinear model	Umbrella model	
101	1,858	1,858	1,858
201	639	337	342
211	514	235	216
103	26	160	164

Fig. 10.38 Umbrella model for spin configuration in CrSe; the spins in each level have a 3-fold symmetry



$a = \sqrt{3}a_0$ and $c = \sqrt{3}c_0$ (Fig. 10.37). To interpret the magnetic reflections in the low temperature pattern, a collinear spin structure was assumed. Table 10.7 shows that the intensities calculated with this model do not agree with observed intensities. A model was then assumed in which the spins were not restricted to being parallel or antiparallel to a single direction. In this structure, planes parallel to the basal plane contain three chromium atoms whose spins form an ‘umbrella’ structure with three-fold symmetry. The intensities calculated from this model are in better agreement with observed intensities. The Shubnikov space group of this structure is $P6_2'm'$ (Fig. 10.38).

10.4.3.7 Helical Structures

Let us first describe a helical structure. Consider a hexagonal crystal consisting of layers perpendicular to the c -axis (Fig. 10.39). The spin moments in the top layers are shown to be along the a -direction. All other spins in this layer will be aligned in the same direction. Consider the next layer. The spins in this layer are parallel to each other but this direction makes angle α_0 with the spin direction in the upper layer. As we go down the c -axis, the direction of spin moments change from one layer to the next by the same angle α_0 . Such a structure is called a ‘helical’ structure. The angle of rotation of the spin from layer to layer is called the ‘turn angle’.

Let us consider the stability of a helical structure. We shall start with the top-most layer in Fig. 10.39. The spins in this layer are all in parallel alignment (ferromagnetic order). We shall denote the exchange integral within the layer by J_0 . We shall denote the exchange integrals for the interaction between the spin in the top layer and a spin in the next layer by J_1 , that with the next by J_2 and so on. The net exchange energy $(E_{\text{ex}})_0$ for the top layer is

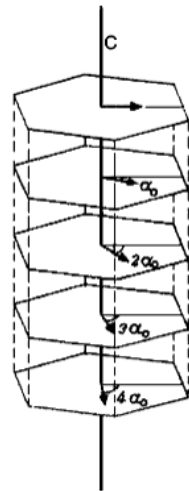
$$(E_{\text{ex}})_0 = -2S^2[J_0 + 2J_1 \cos \alpha_0 + 2J_2 \cos 2\alpha_0 + \cdots] \quad (10.13)$$

where S is the spin. For N layers, the energy is

$$E_{\text{ex}} = -2S^2N[J_0 + 2J_1 \cos \alpha_0 + 2J_2 \cos 2\alpha_0 + \cdots]. \quad (10.14)$$

It can be shown that this energy has a minimum for some value $(\alpha_0)_{\text{min}}$ of the turn angle i.e. the helical structure is stable when the turn angle is $(\alpha_0)_{\text{min}}$. Enz [21] showed that for a 3-layer system

Fig. 10.39 Helical spin configuration in a hexagonal crystal



$$(\alpha_0)_{\min} = -J_1/4J_2. \quad (10.15)$$

In general, $(\alpha_0)_{\min}$ has values other than 0 and π .

We shall now discuss the first helical structure that was observed viz., the magnetic structure of MnAu_2 . MnAu_2 is antiferromagnetic with $T_N \approx 363$ K. The chemical crystal structure of MnAu_2 is shown in Fig. 10.40a wherein the Mn positions are shown. X-ray studies showed that MnAu_2 has the tetragonal body-centred arrangement of Mn atoms with $c/a = 2.6$. Each Mn atom has 2 Au atoms at the same distance above and below the Mn atom. The Mn atoms are in layers perpendicular to the c -direction, the interlayer distance being $c/2$.

The neutron diffraction patterns of MnAu_2 recorded by Herpin and Meriel [22] at 363 and 293 K are shown in Fig. 10.41a, b. The (002) and (101) peaks observed at 363 K are due to chemical (nonmagnetic) state due to nuclear scattering. These peaks are also present in the LT pattern. But there are two extra peaks on either side of (002). Also, the (101) peaks appear to have broadened. By taking the difference of a and b, the broad peak (101) also shows up two peaks on either side of (101) as can be seen in Fig. 10.41c. The presence of these peaks could be explained by a helical arrangement of spin moments along the c -axis with a turn angle of $\alpha_0 = 51^\circ$ per layer (or $102^\circ/\text{unit cell}$) along c . This spin arrangement is shown in Fig. 10.40b.

Helical magnetic structures have been observed in MnO_2 , CrSb , CuCr_2O_2 and MnCr_2O_4 .

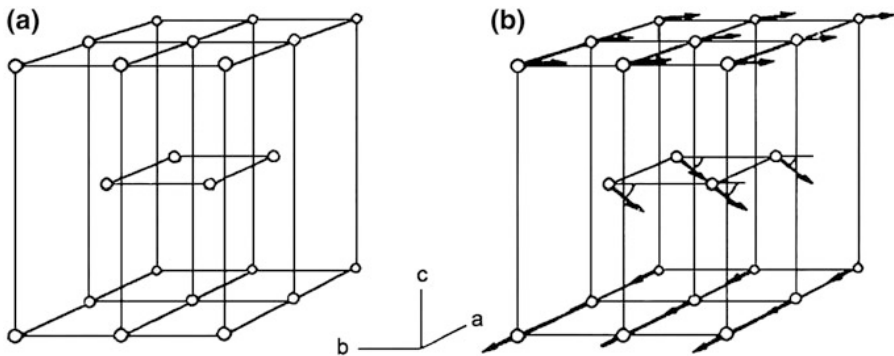
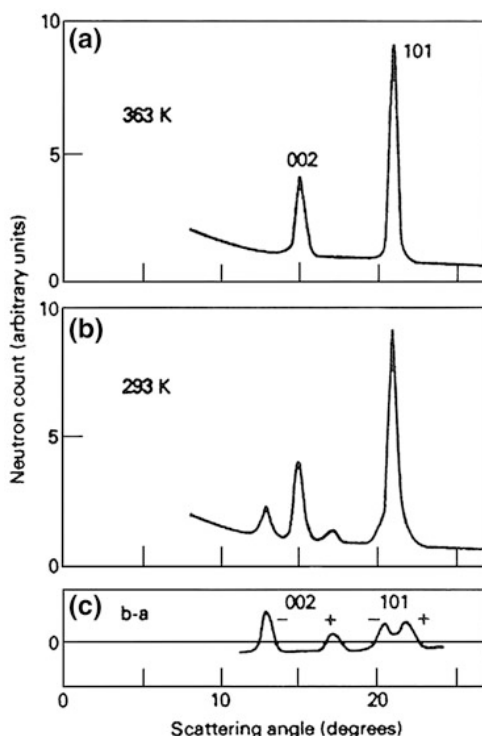


Fig. 10.40 **a** Chemical structure of MnAu_2 , **b** Helical magnetic structure of MnAu_2 (only Mn positions are shown)

Fig. 10.41 Neutron diffraction patterns for MnAu_2
a At 363 K and **b** At 293 K,
c Difference of **(b)** and **(a)**



10.4.4 Ferrimagnetic Crystals

10.4.4.1 Magnetite

Magnetite (Fe_3O_4) is a ferrite with the spinel structure. One fourth of the unit cell is shown in Fig. 10.42. Half the Fe^{3+} ions are located at the tetrahedral sites. The octahedral sites are occupied by all the Fe^{2+} ions and the remaining half of the Fe^{3+} ions. Magnetic susceptibility measurements show that Fe_3O_4 is ferrimagnetic with $T_{\text{FN}} = 858 \text{ K}$.

X-ray diffraction and neutron diffraction patterns at RT recorded by Shull et al. [16] are shown in Fig. 10.43. The chemical structure is the same in both patterns. From the nuclear positions known from X-ray structure determination, we can calculate the nuclear scattering contribution to intensities using nuclear scattering amplitudes. Similarly, based on an assumed model of spins, we can calculate the contribution of magnetic scattering using appropriate magnetic form factors. The sum of the two contributions is the calculated intensity for a reflection.

The ferrimagnetic structure assumed for Fe_3O_4 is shown in Fig. 10.44. The Fe ions in the tetrahedral positions have their spin moments pointed upwards whereas those in the octahedral sites have spin moments pointed downwards. These being unequal, the substance behaves like a ferrimagnet. The calculated and observed

Fig. 10.42 Chemical structure of Fe_3O_4 (one fourth of unit cell)

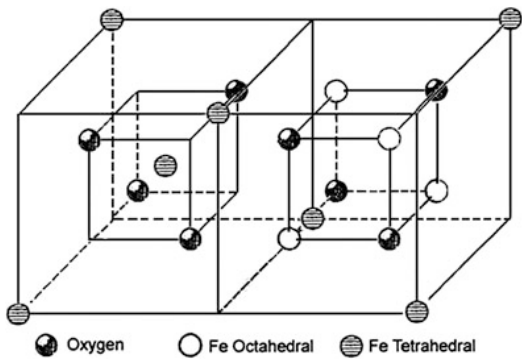


Fig. 10.43 X-ray and neutron diffraction patterns for Fe_3O_4

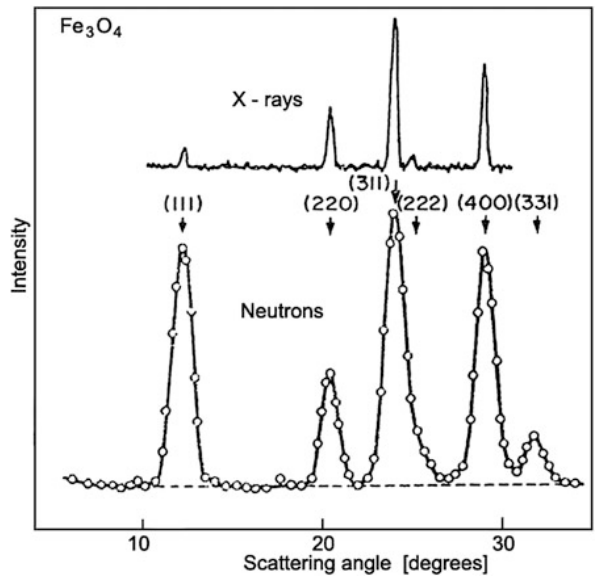


Fig. 10.44 Magnetic structure of Fe_3O_4 (one fourth of unit cell)

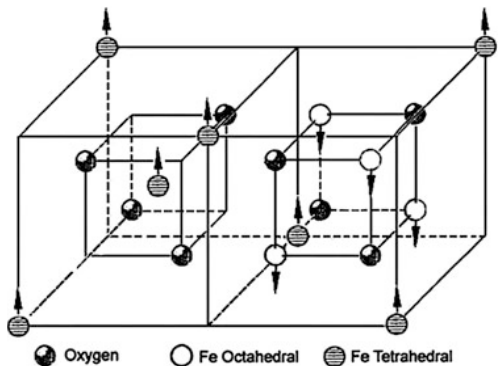


Table 10.8 Comparison between observed and calculated intensities for Fe₃O₄

Reflection	Calculated intensities			Observed intensity
	Magnetic	Nuclear	Total	
(111)	902	32	934	860
(220)	125	218	343	360
(311) } (222) }	112	948	1060	1070
(400)	116	649	765	780
(331)	94	116	110	135
(422) } (333) } (511) }	20	670	690	700
(440) } (531) }	32	1692	1724	1730

intensities are given in Table 10.8. The agreement is satisfactory. This model was adopted after other models were found to give unsatisfactory intensities.

10.4.4.2 Copper Chromite

Copper chromite (CuCr₂O₄) is tetragonal with a slightly distorted spinel structure (Fig. 10.45). The divalent Cu ions are at the A sites and the trivalent Cr ions are at the B sites. Magnetic susceptibility measurements show that the compound is ferrimagnetic with a Neel temperature $T_{\text{FN}} = 135$ K and a magnetic moment of $\sim 0.4 \mu_{\text{B}}$ /molecule.

The neutron diffraction patterns obtained by Prince [23] at RT and LN temperature are shown in Fig. 10.46. The low temperature pattern shows the (101) and (002) reflections which are purely magnetic. The magnetic reflections as well as the nuclear reflections can be indexed on the basis of the chemical cell. On the other hand, the (002) reflection which is present with unmistakable intensity is not permitted by the space group of the chemical cell because of the *d*-glide operation. The appearance of this reflection can be explained as due to nonparallel spins of Cr ions related by the glide operation. There is no contribution of magnetic scattering to the (110) and (112) reflections which means that Cr ions in the same sheet have parallel spins. These facts indicate that the Cr ions in any (001) sheet are parallel and those in adjacent sheets make angles with each other. The $\sim 0.5 \mu_{\text{B}}$ magnetic moment of a molecule can be explained by assuming one of the two possible arrangements of the Cr ions shown in Fig. 10.47, two ions having moment of $3 \mu_{\text{B}}$ and one ion having a moment of $1 \mu_{\text{B}}$. Thus the spin configuration in copper chromite crystal is ferrimagnetic.

Fig. 10.45 CuCr_2O_4 structure projected on plane perpendicular to $[001]$

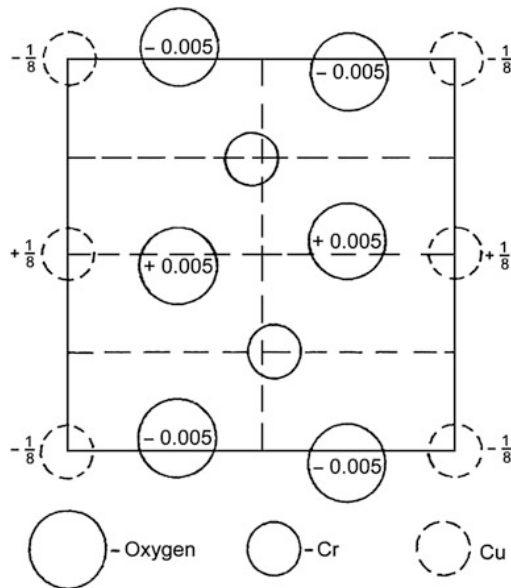
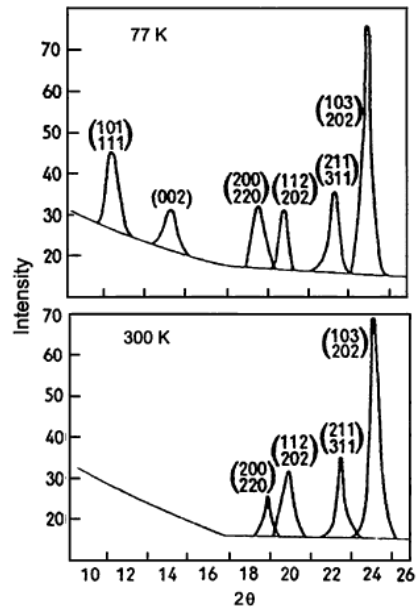


Fig. 10.46 Neutron diffraction patterns for CuCr_2O_4 at 300 and 77 K



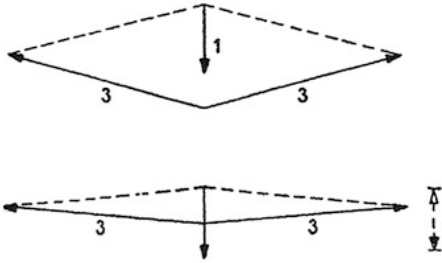
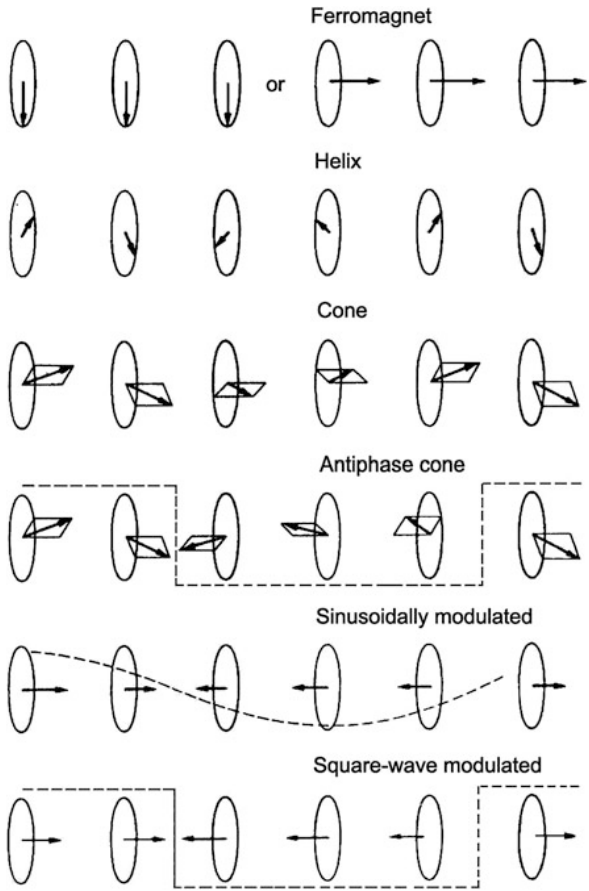


Fig. 10.47 Two possible ways of arranging moments of Cr ions to yield observed moment of $0.5 \mu_B/\text{molecule}$

10.4.5 Rare Earth Metals

Although the rare earth metals have the same electronic structure (unpaired electrons in the $4f$ shell) they display a variety of exotic magnetic structures. At a given

Fig. 10.48 Variety of magnetic structures found in rare earths



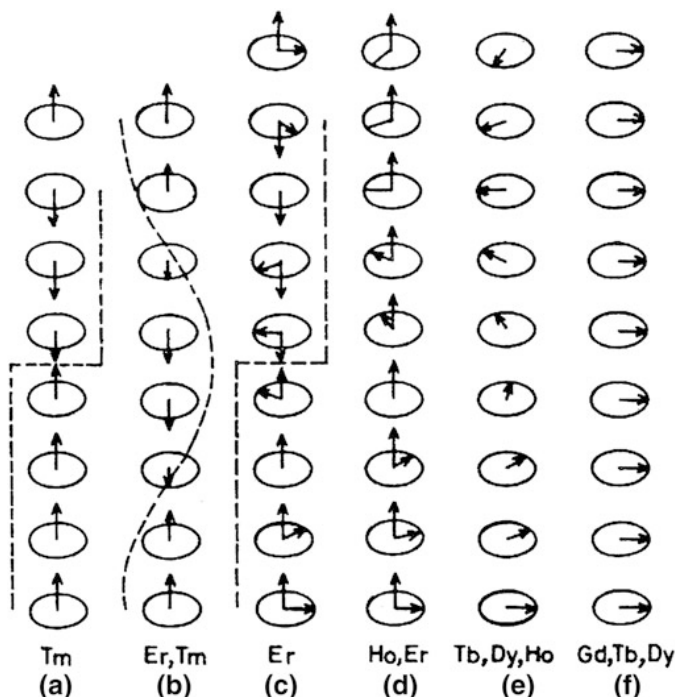


Fig. 10.49 Magnetic structures of specific rare earth metals

temperature, different rare earths have different magnetic structures. Further, a given rare earth metal has different magnetic structures in different temperature ranges. A detailed discussion of the magnetic structures of rare earth metals is given by Koehler [24], Kubo and Nagamiya [25] and Sinha [26]. Since we are now well-acquainted with the procedure of determining the magnetic structure, we shall only summarize the results on magnetic structures of rare earth metals.

The different types of magnetic structures found in the rare earth metals are shown in Fig. 10.48. The magnetic structures assumed by specific rare earth metals are shown in Fig. 10.49. The observations on magnetic structures of rare earth metals are summarized in Table 10.9.

Tm (Fig. 10.49a) has a seven-layer repeat distance (3 layers of north-pointing moments and 4 layers of south-pointing moments). An oscillatory moment is observed in Er (Fig. 10.49b) in which the z -components of the moments in a hexagonal layer are parallel to each other and the magnitude of this component varies sinusoidally. At 53 K, Er displays type c (Fig. 10.49), where several layers with moments pointing up are followed by an equal number of layers with moments pointing down. Ho and Er also display type d (Fig. 10.49) in which the perpendicular component of the moment shows a helical variation. Tb, Dy and Ho display the helical structure shown in Fig. 10.49e. At different temperature ranges, Gd, Tb and Dy transform to the ferromagnetic structure (Fig. 10.49f).

Table 10.9 The magnetism and crystal structure of rare-earth metals (μ_r is magnetic moment)

Metal	Crystal structure	Magnetism	Remarks
Ce	Hexagonal ABAC type ($c/2a = 1.63$); face-centred cubic	A hexagonal ABAC type is antiferromagnetic with $T_N = 12.5$ K	Highly dense nonmagnetic face-centred cubic structure (no f electrons)
Pr	Hexagonal ABAC type ($c/2a = 1.62$)	Antiferromagnetic with $T_N = 25$ K. Sine structure in b_1 direction	Either the spins on the BC planes are ordered, or those on the A, A planes
Nd	Hexagonal ABAC type ($c/2a = 1.61$)	Antiferromagnetic with $T_N = 19$ K. Sine structure in b_1 direction	Above 7.5 K only the B, C-plane spins are ordered; below 7.5 K those in or those on A, A are ordered also
Gd	Hexagonal close-packed ($c/a = 1.59$), the coefficient of expansion c below T_c is large if negative, whereas a is small if positive	Ferromagnetic, $T_c = 289$ K 289–245 K: $\mu // c$ axis; 245–225 K: continuous change from $\mu // c$ to $\mu \perp c$ 225–165 K: $\mu \perp c$ axis; 165–0 K: continuous change from $\mu \perp c$ to angle $\angle(\mu, c) = 34^\circ$	
Tb	Hexagonal close-packed ($c/a = 1.58$)	Spiral structure from $T_N = 228$ to 221 K and $\mu \perp c$; ferromagnetic below 221 K; $\mu // b$ to μ (0 K) = $9.34 \mu_B$	
Dy	Hexagonal close-packed ($c/a = 1.57$), $c/a = 1.59$ at low temperatures	Spiral structure from $T_N = 179$ to 85 K, $\mu \perp c$; ferromagnetic below 85 K; $\mu // a, \mu$ (0 K) = $10.0 \mu_B$	
Ho	Hexagonal close-packed ($c/a = 1.57$)	Spiral structure from $T_N = 132$ to 20 K, $\mu \perp c$; conic structure below 20 K; moment in the c axis direction $117 \mu_B$; rotational component perpendicular to the axis $10.2 \mu_B$	
Er	Hexagonal close-packed ($c/a = 1.57$), at low temperatures $c/a = 1.575$	Sine structure parallel to the c axis from $T_N = 85$ to 53 K; components parallel to and perpendicular with the c axis vibrate from 53 K to 20 K; conic structure below 20 K; $\mu // c = 8, \mu$ (0 K) = $\mu_B, \mu \perp c = 4 \mu_B$	
Tm	Hexagonal close-packed ($c/a = 1.57$)	Sine structure parallel to the c axis below $T_N = 56$ K; gradually start vibrating circularly from 40 K down, then ferrimagnetic with ++++---- as the period; each atom has μ (0 K) = $7 \mu_B$	

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